

Inverse Problems in Geophysics

Exercise 3: Problem types

2. MGPY+MGIN

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Task linear regression

1. Set up a synthetic model, create data and add noise
2. Compute the least-squares solution
3. Plot the model response along with the data
4. Plot misfit and compare it with noise
5. Compute the data information matrix and plot it
6. Plot the data importance (diagonal) over x
7. Add an outlier, redo inversion and use IRLS

Error weighting and outliers

replace d_i by $\hat{d}_i = d_i/\epsilon_i$ and $\hat{\mathbf{G}} = \text{diag}(1/\epsilon_i) \cdot \mathbf{G} \Rightarrow$

$$\mathbf{m} = (\hat{\mathbf{G}}^T \hat{\mathbf{G}})^{-1} \hat{\mathbf{G}}^T \hat{\mathbf{d}}$$

robust inversion - **The pragmatic way**

- do inversion and look at misfit distribution
- delete all data with high misfit (or set large error)
- repeat

Error weighting and IRLS

replace d_i by $\hat{d}_i = d_i/\epsilon_i$ and $\hat{\mathbf{G}} = \text{diag}(1/\epsilon_i) \cdot \mathbf{G} \Rightarrow$

$$\mathbf{m} = (\hat{\mathbf{G}}^T \hat{\mathbf{G}})^{-1} \hat{\mathbf{G}}^T \hat{\mathbf{d}}$$

Iteratively Reweighted Least-Squares (IRLS) - robust inversion

$$w_i = \frac{|v_i|/\|\mathbf{v}\|_1^1}{v_i^2/\|\mathbf{v}\|_2^2} = \frac{\|\mathbf{v}\|_2^2}{|v_i| \cdot \|\mathbf{v}\|_1^1}$$

$$\epsilon_i \Rightarrow \epsilon_i/\max(w_i, 1)$$

Problem types

Example linear problem: Traveltime tomography

$$t = \int_L \frac{1}{v} dl = \int_L s dl$$

piece-wise constant s : $t = \sum_j s_j \Delta l_j \Rightarrow t_i = \sum W_{ij} s_j$

$\mathbf{d} = \mathbf{W}\mathbf{m}$ where $\mathbf{W}$ contains lengths of rays (rows) in cells (columns)

with $\mathbf{d} = [t_1, \dots, t_N]$, $\mathbf{m} = [s_1, \dots, s_M]$

Types of inverse problems

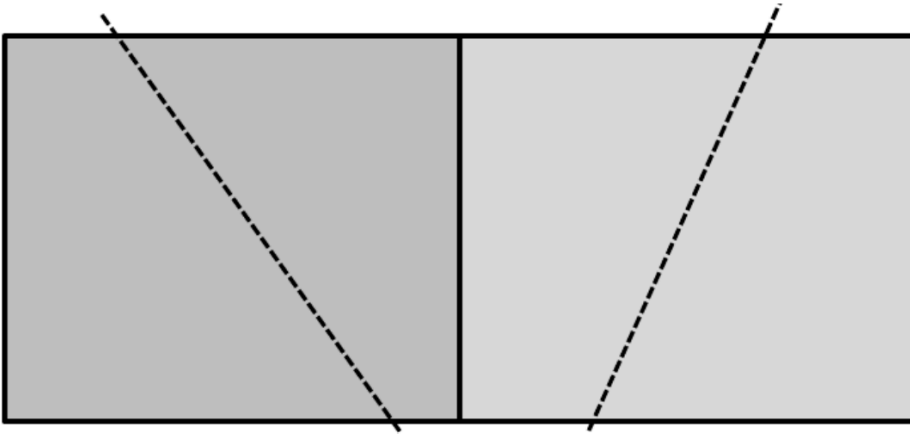
$$\textcircled{!} \mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n} \Rightarrow \mathbf{G} \in \mathbb{R}^{N \times M}$$

rows stand for measurements, columns represent model parameters

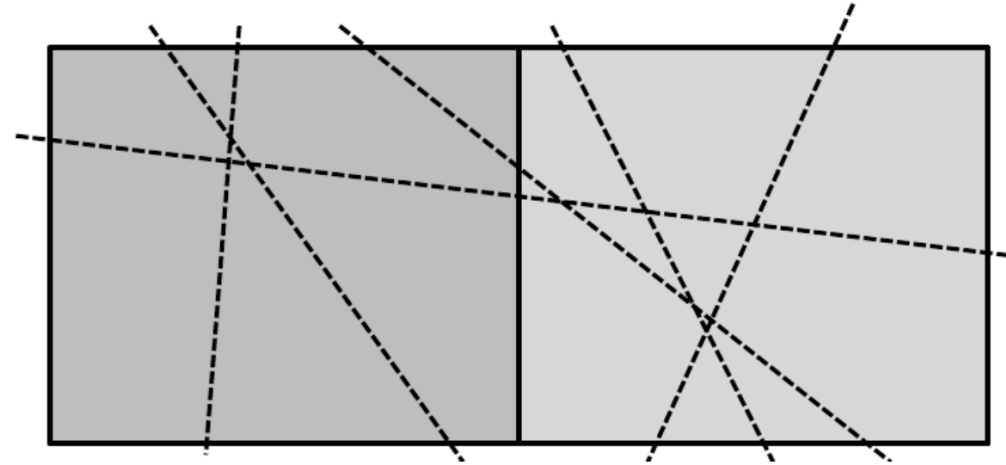
- $N = M$: even-determined
- $N > M$: over-determined problem (no existing solution)
- $N < M$: under-determined problem (no unique solution)
- mostly: (both over- and) under-determined model parts

Types of inverse problems (Menke, 2012)

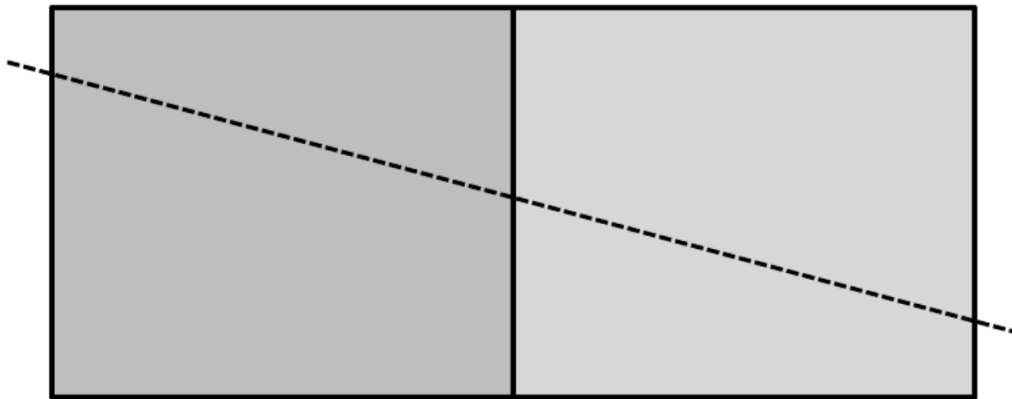
Even-determined



Over-determined

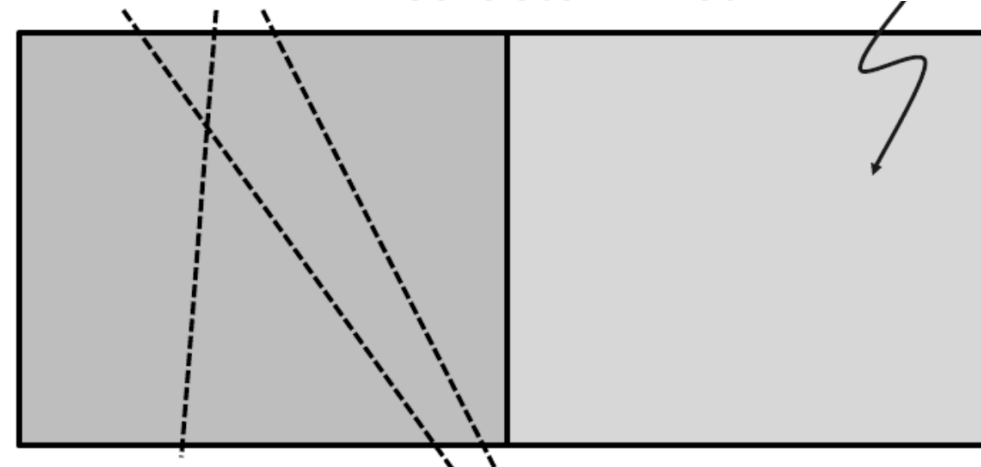


Under-determined

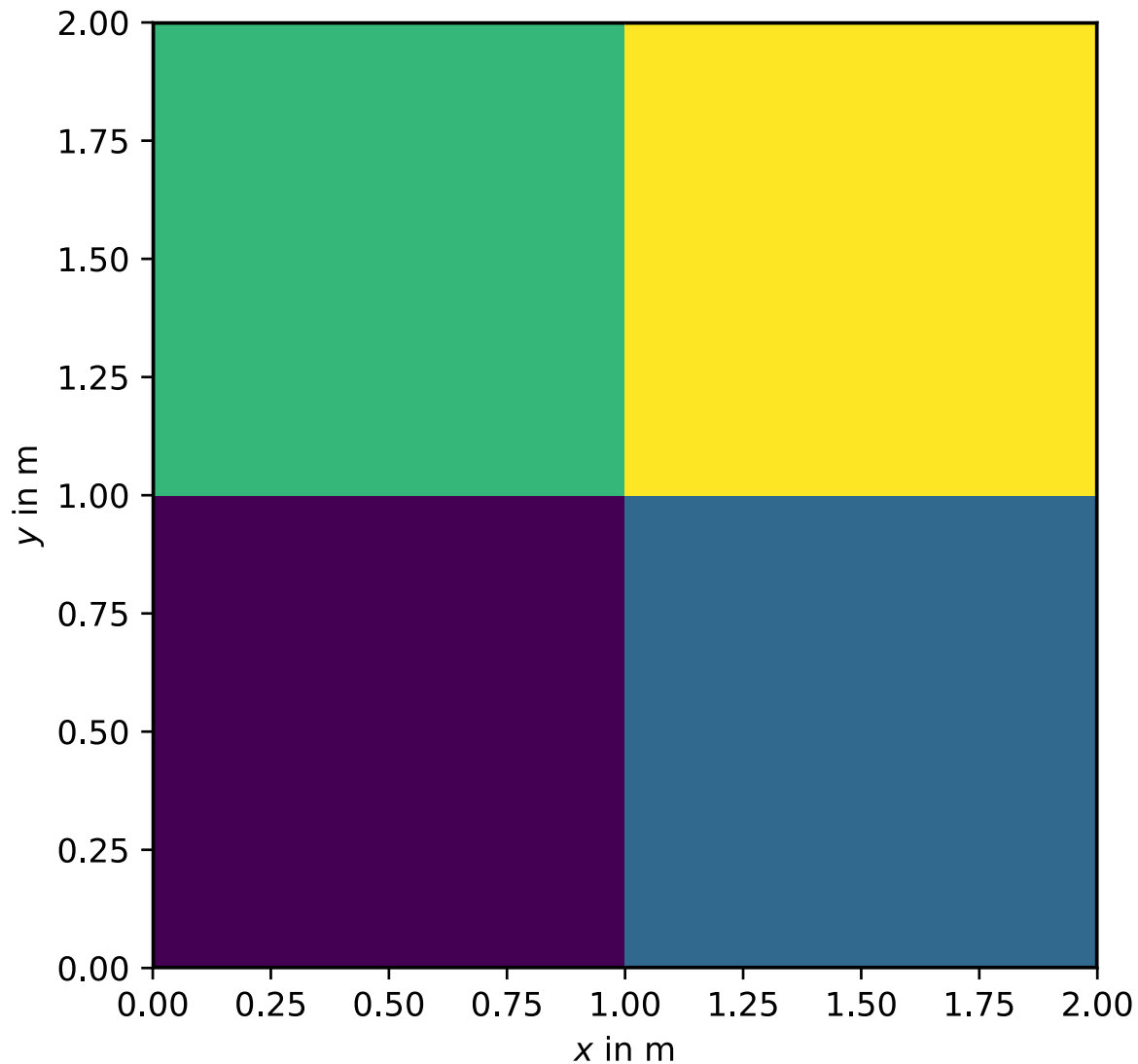


Mixed-determined

null space

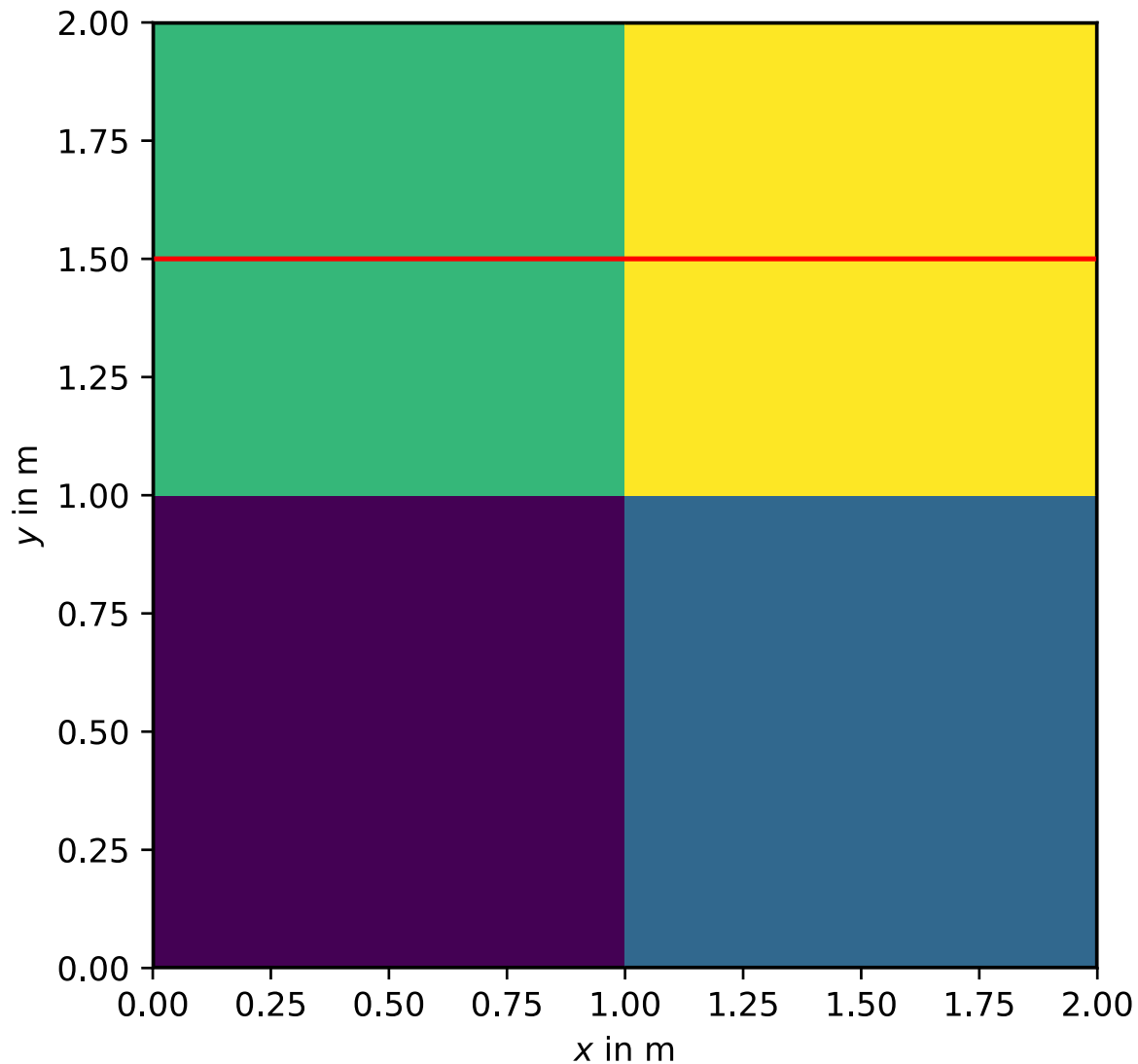


Problem types - model



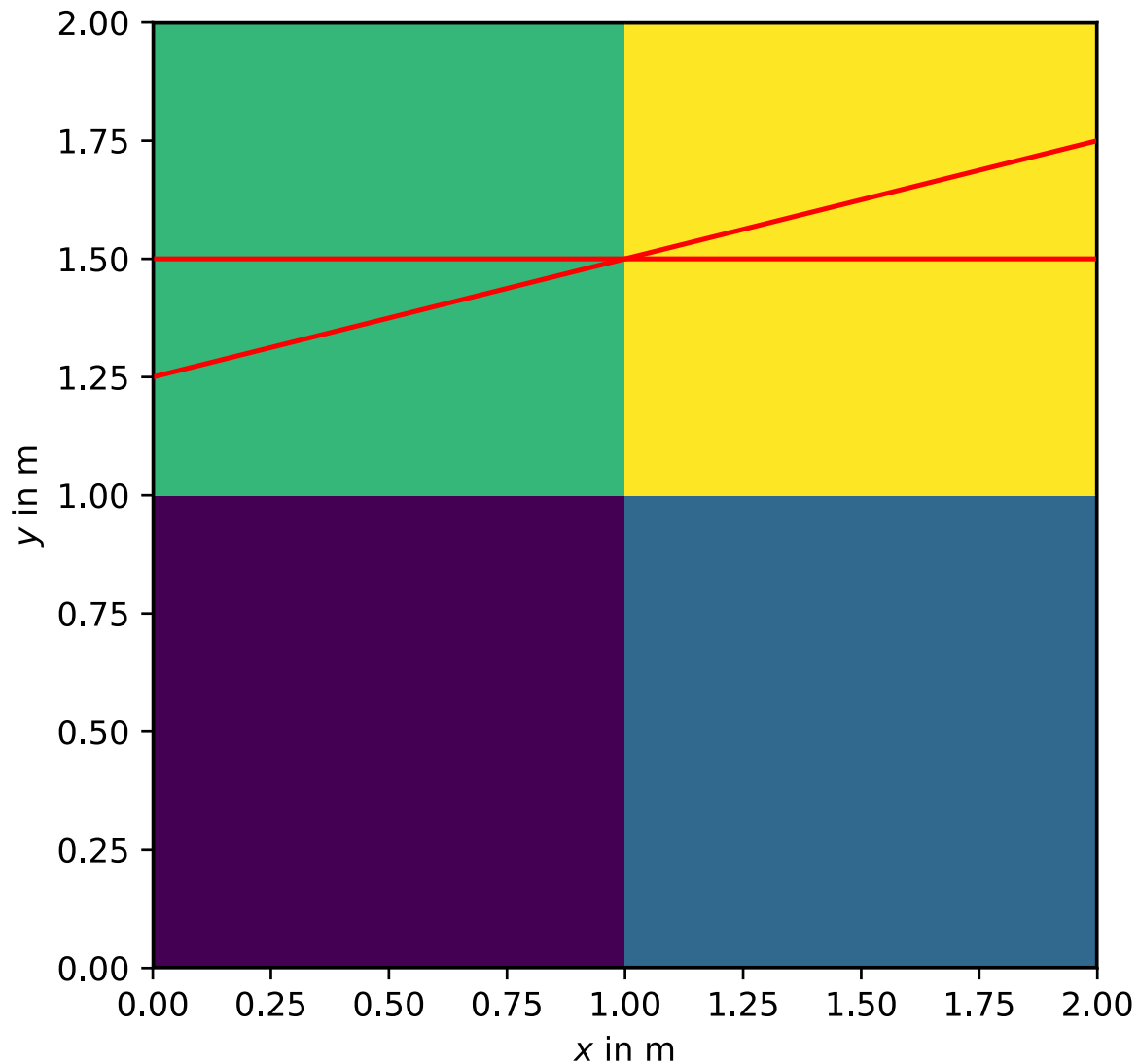
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 1



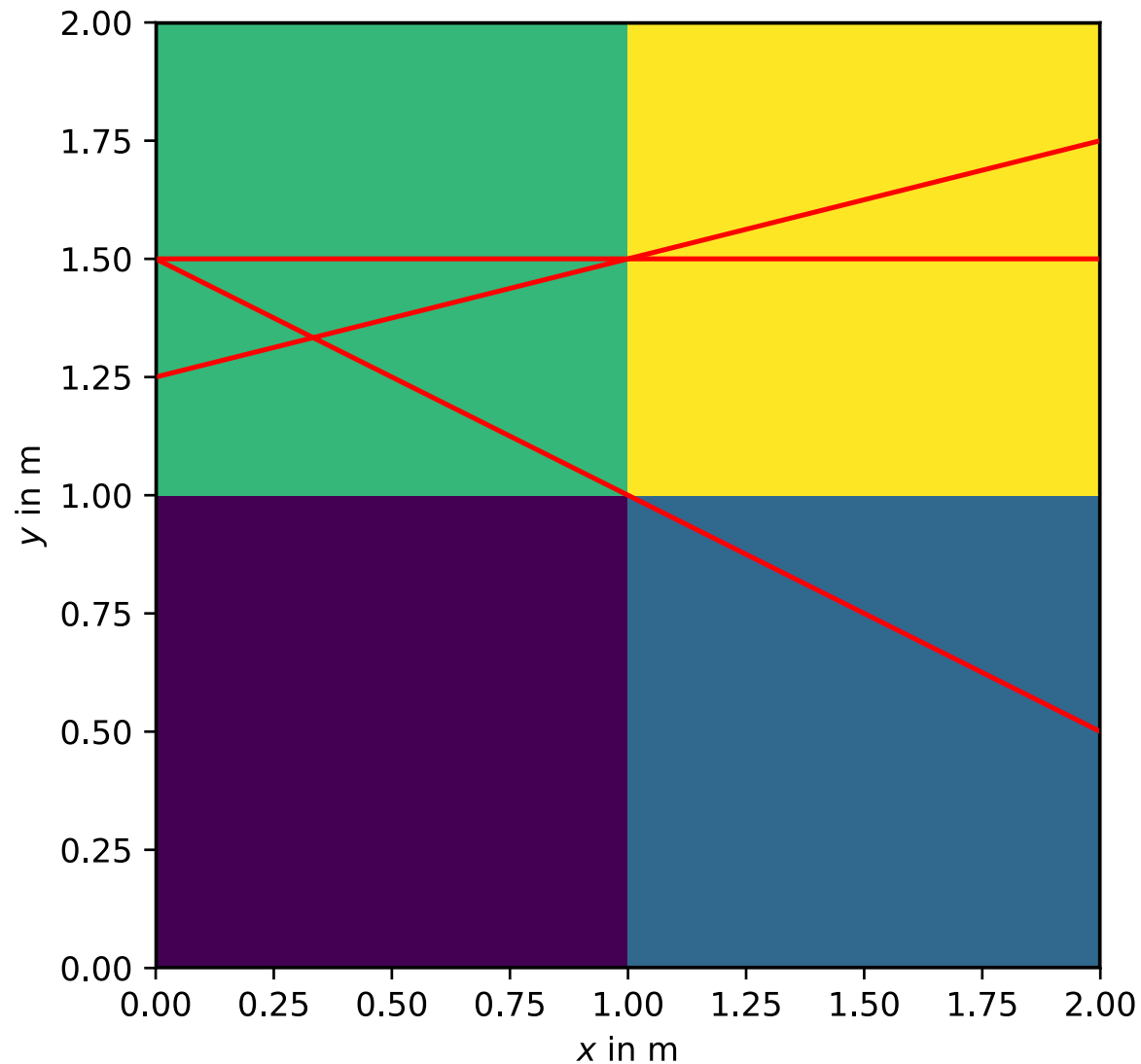
- guess whether it is a) over-, b) under-, or c) mixed-determined
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Problem types - 2



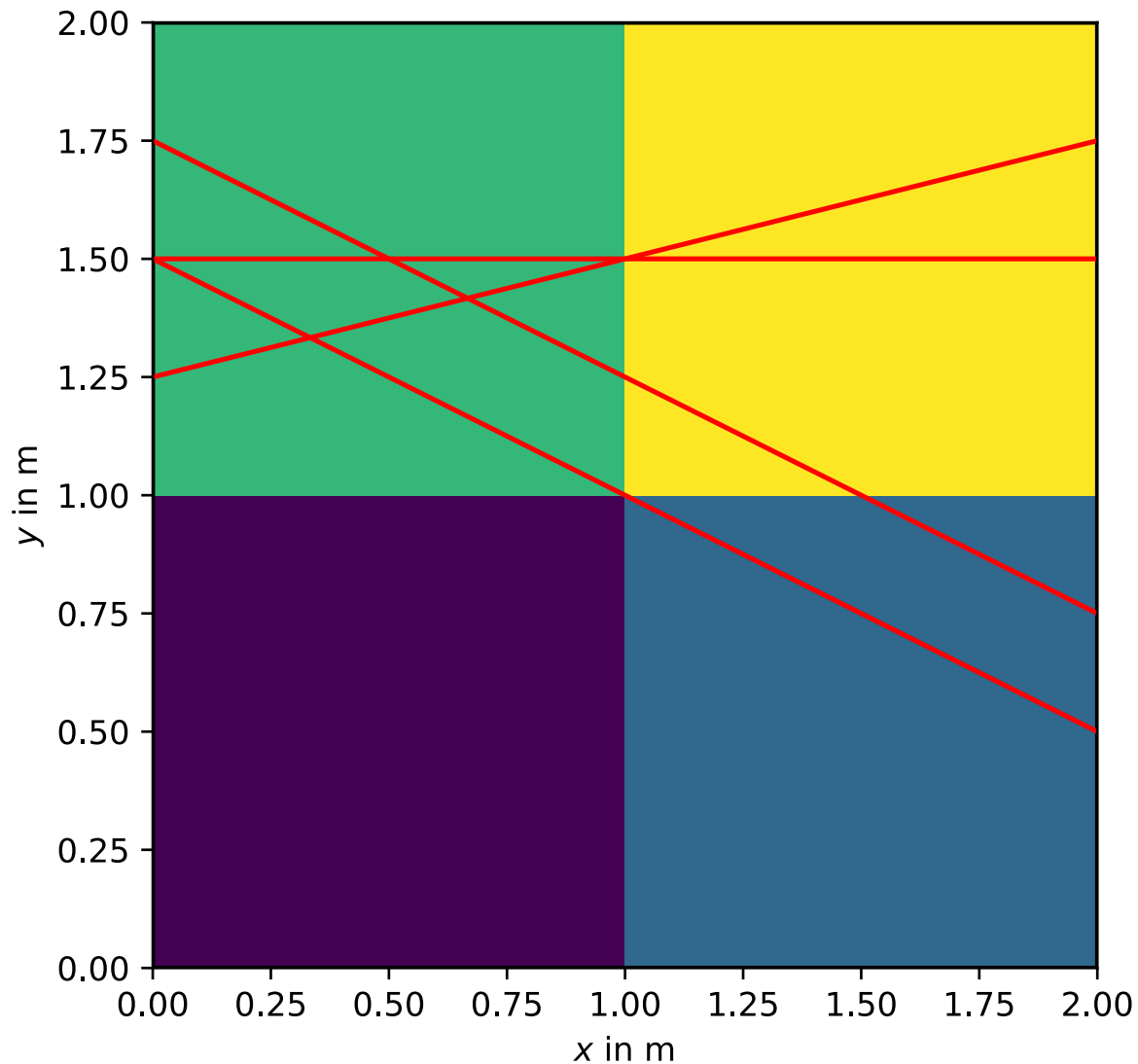
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 3



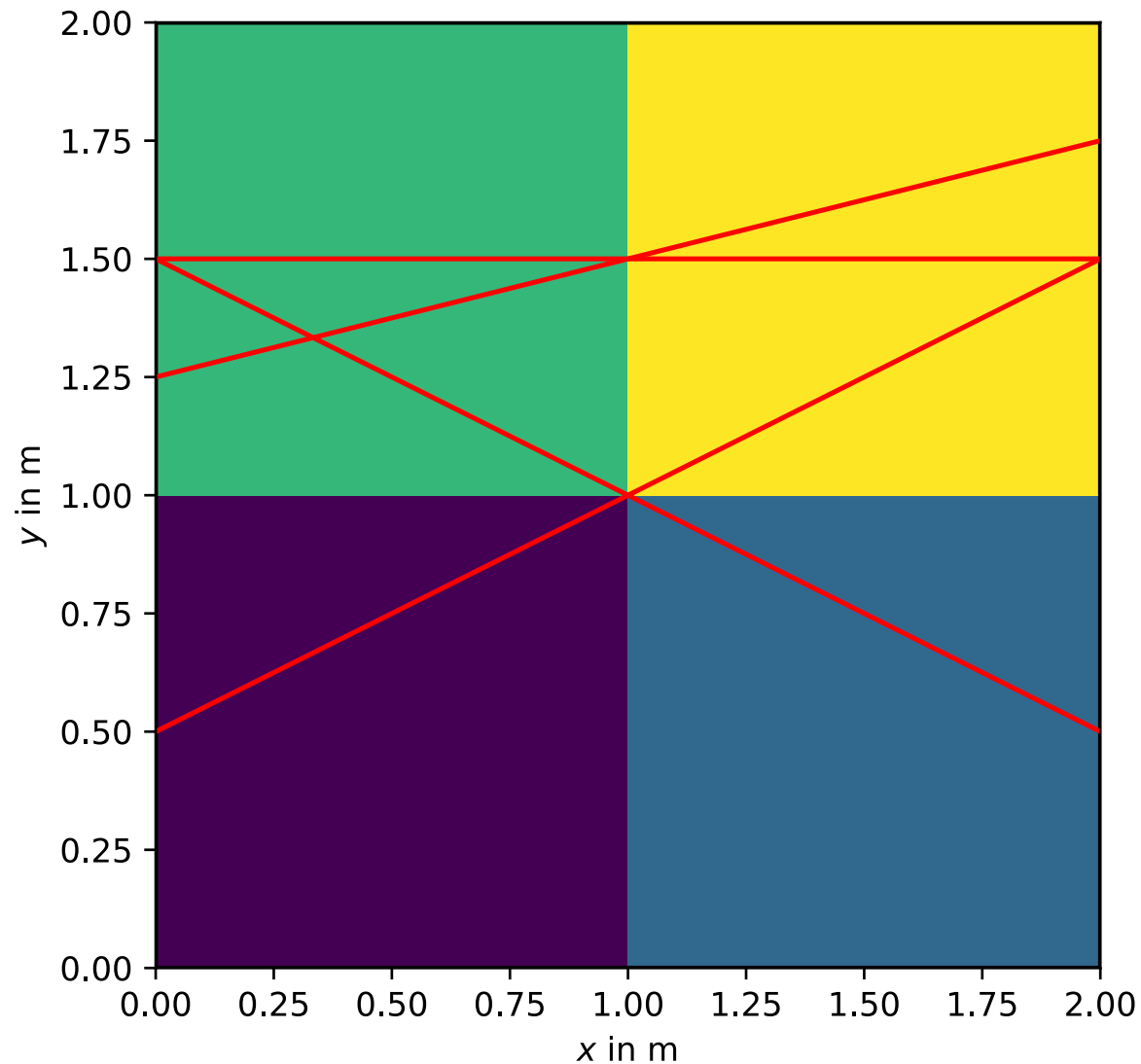
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 4



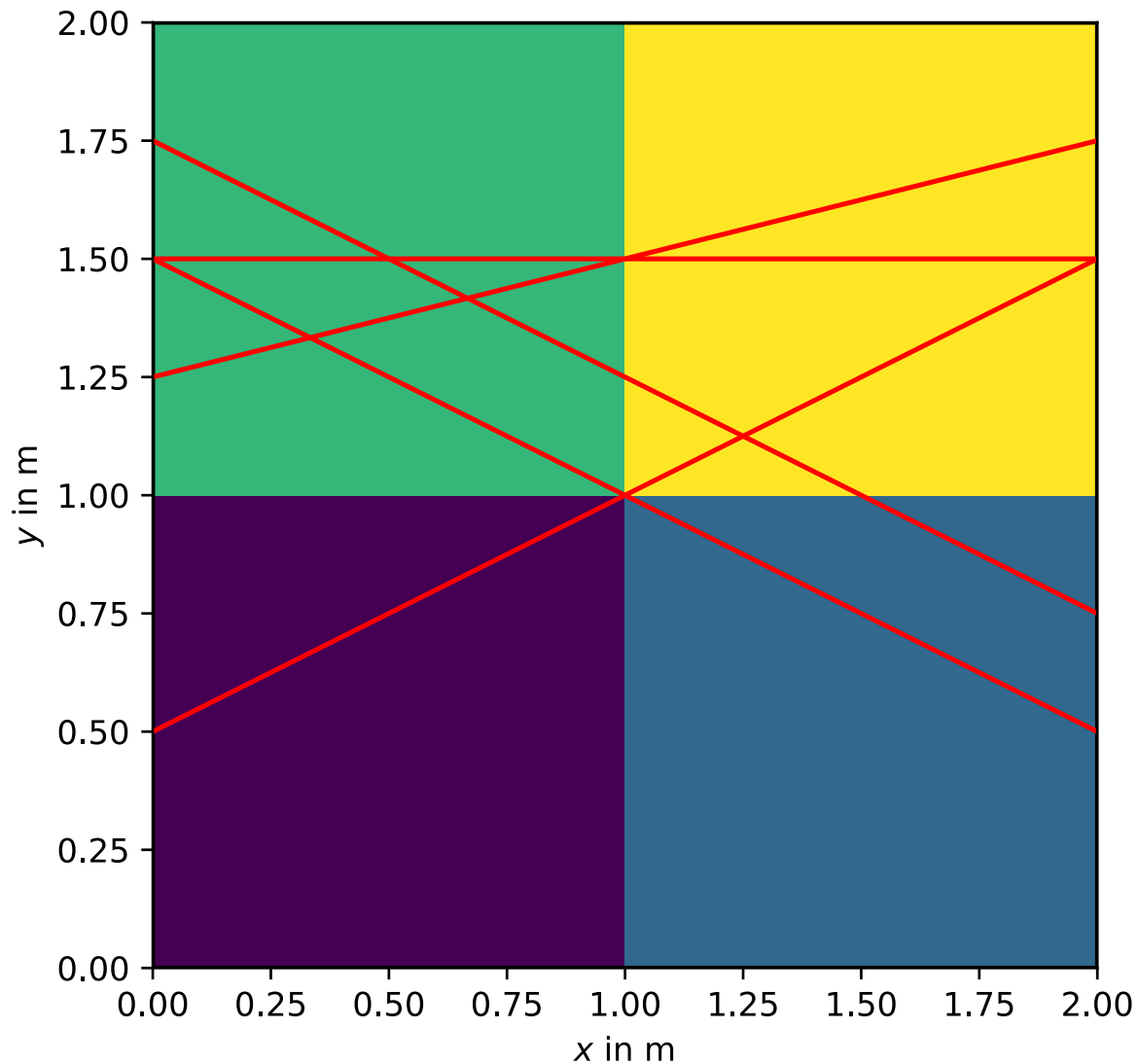
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 5



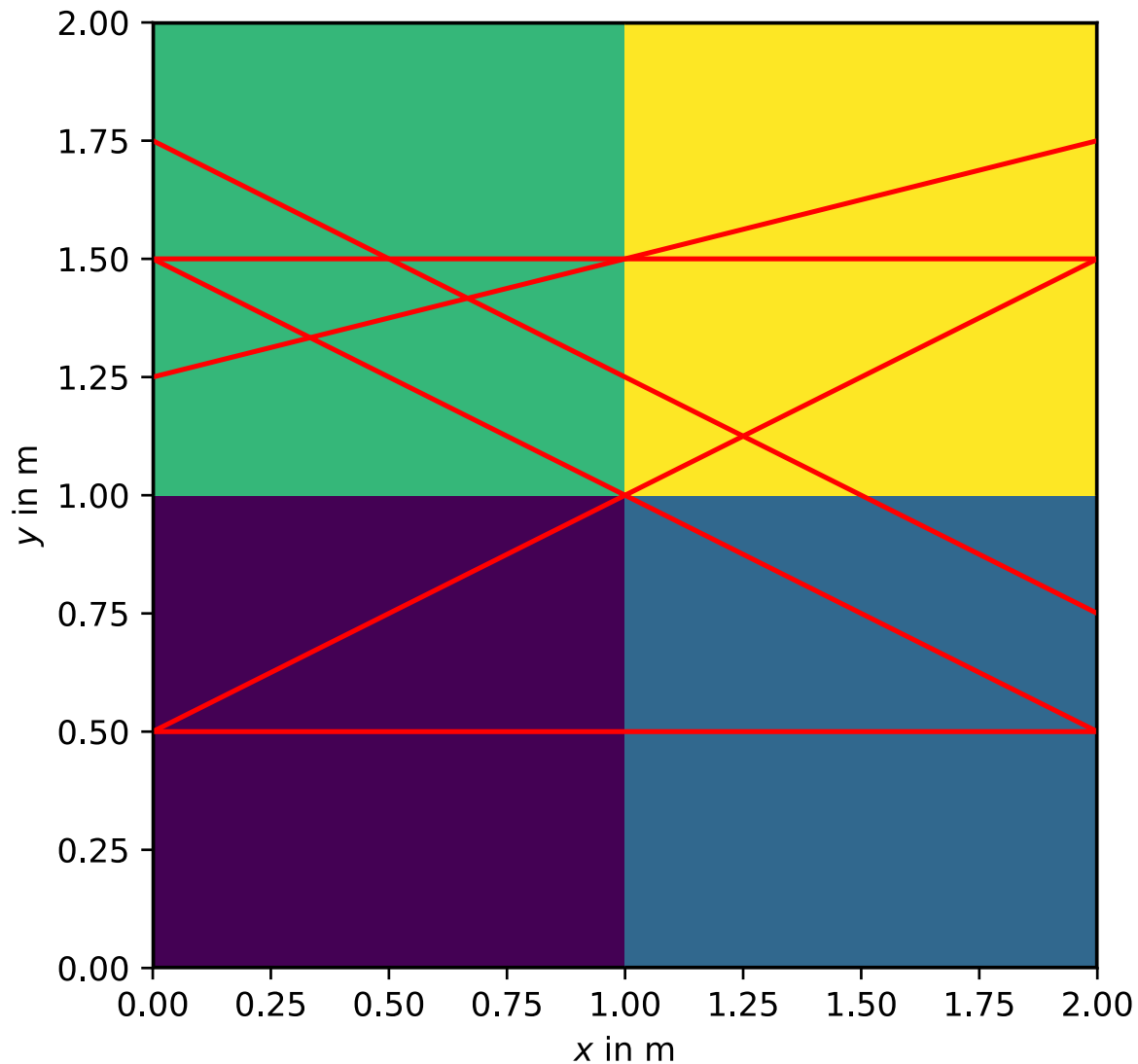
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 6



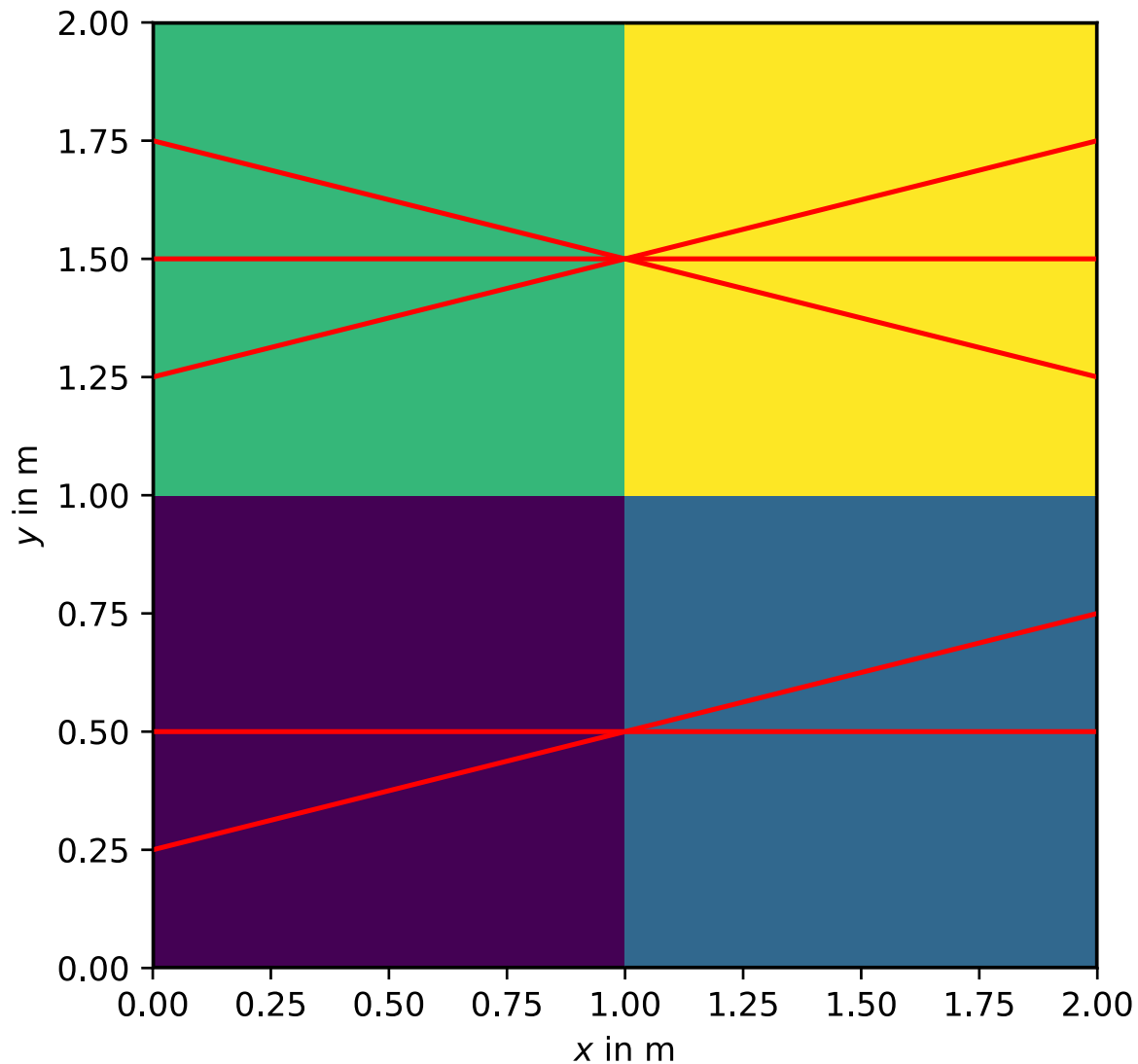
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 7



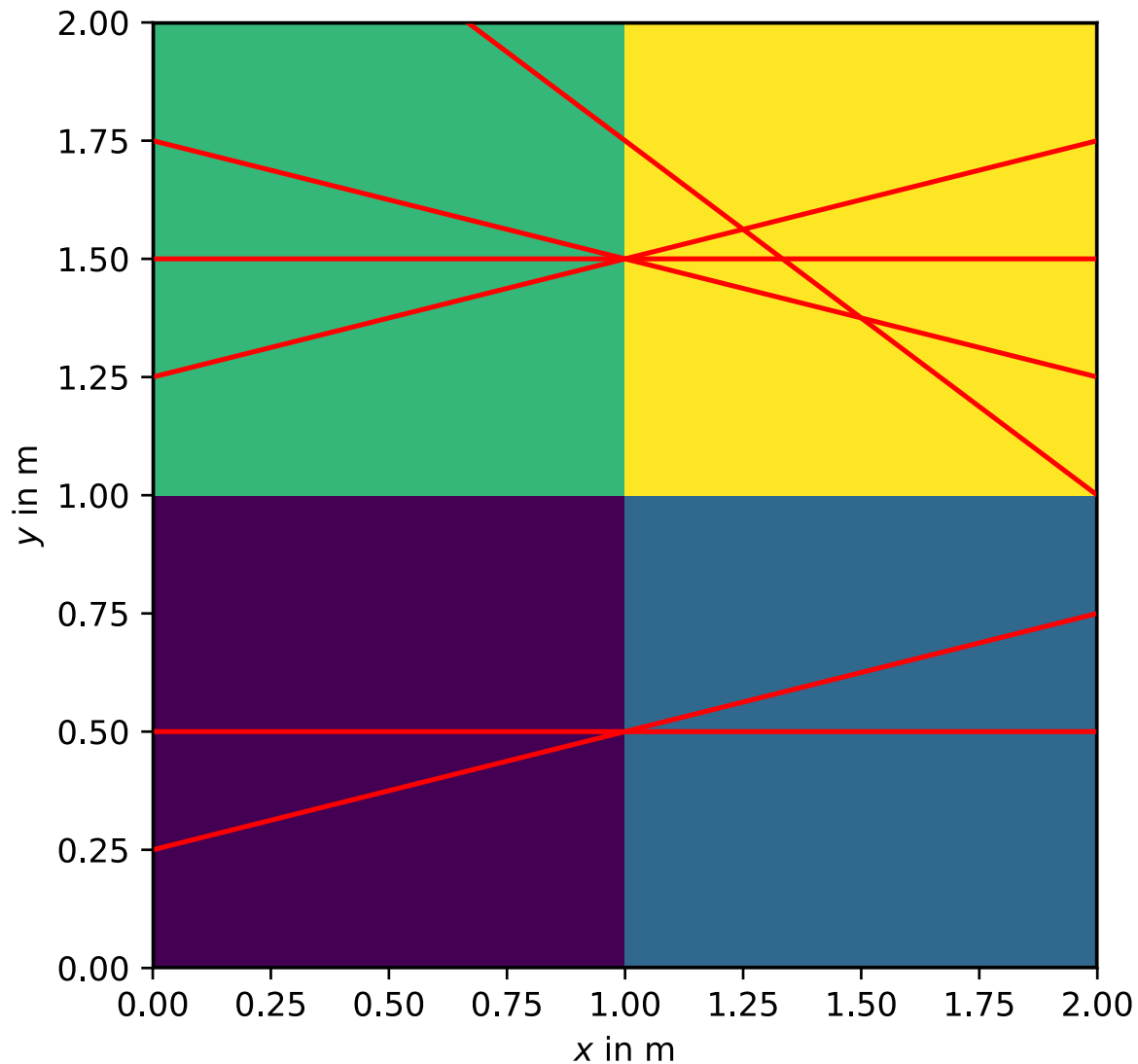
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 8



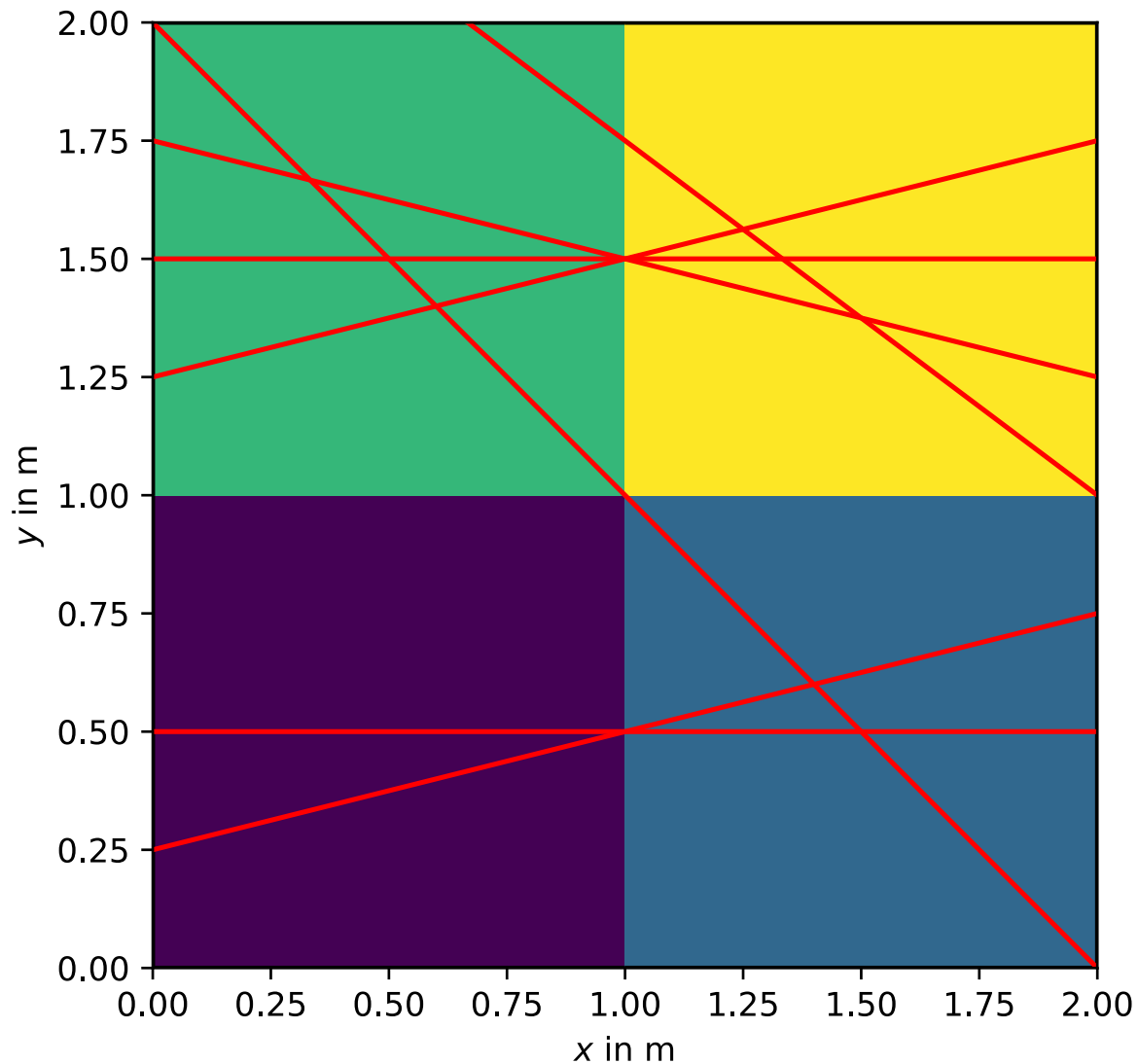
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 9



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

Problem types - 10



- guess a) even, b) over-, c) under-, or d) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

